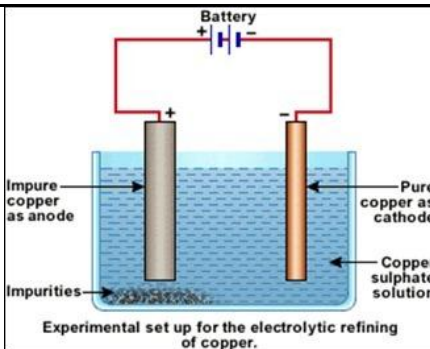


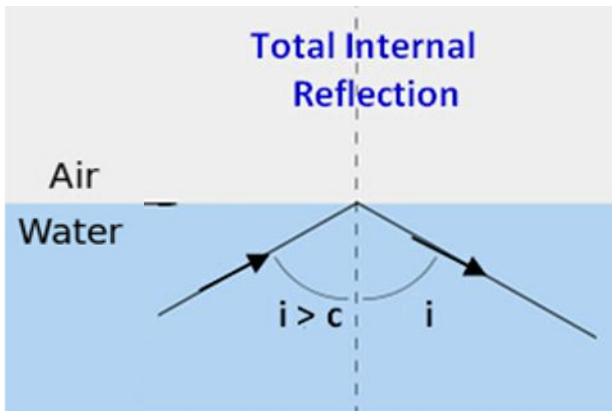
SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुञ्जिका

यस कुञ्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ ।
 उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नुपर्नेछ, तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

प्र. नं.	उत्तर	अंक
खण्ड 'क' (Section 'A')		
1 a.	(iii) Digital	1
b	(ii) Embryo will develop into a male because of Y-chromosome	1
c	(ii) a-ii, b-iii, c-i	1
d	(iii) 2 second	1
e	(iii) $W_3 < W_2 < W_1$	1
f	(i) The image is virtual and magnified.	1
g	(iii) Dynamo	1
h	(iii) carbon-carbon has single bond and general formula C_nH_{2n+2}	1
i	(ii) Alkyl benzene sulphonate	1
j	(iv) Mixing with oil	1
खण्ड 'ख' (Section 'B')		
2.a	- The dependent variable is V (Potential difference)	1
b	- RBC helps to carry oxygen from lungs to the different parts of the body.	1
c	- Establishment of conservation area, Implementation of strict rules and regulations	1
d	- Pascal's law	1
e	- 3	1
f	- Roasting is done in the presence of excess of air and calcination is done in the absence or limited supply of air or roasting is done for sulfide ore and calcination is done for carbonate ore - Or any other correct answer will get 1 mark.	1
g	 Experimental set up for the electrolytic refining of copper.	1
h	- In Vitro Fertilization (IVF) is a boon for childless couples by providing a highly effective pathway to parenthood when natural conception is not possible due to biological barriers like blocked fallopian tubes, low sperm count, etc.	1

2.i	- Use Rittha or ash as cleansing agents	1				
	खण्ड 'ग' (Section 'C')					
3	- Substituting the related units - LHS = S =m - RHS= $ut+1/2at = ms^{-1}s + 1/2ms^{-2}s = m + \frac{1}{2} ms^{-1}$ - The units on LHS and RHS are not equal so the given equation is not valid.	1 1				
4	<table><tr><th>Bat</th><th>Pigeon</th></tr><tr><td>1. Bat is viviparous. 2. Bat is a terrestrial animal.</td><td>1. Pigeon is oviparous. 2. Pigeon is an aerial animal.</td></tr></table>	Bat	Pigeon	1. Bat is viviparous. 2. Bat is a terrestrial animal.	1. Pigeon is oviparous. 2. Pigeon is an aerial animal.	1 1
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5	(i) Sub-division of A is Pteridophyta (ii) Plant B bears seed but not in A.	1 1				
6	X = Fertilization, Y = Parthenogenesis	1 1				
7	- Artificial insemination - Advantage: increases the rate of fertilization, collected semen can be used even after the death of the male Or any other correct answer will get 1 mark.	1 1				
8	- High amount of carbon emission from vehicles or industries - Volcanic eruptions - Any one correct/ two correct reasons get 1 mark	1 1				
9	Formula, $g = \frac{GM}{R^2}$, $R = 7.1 \times 10^7 m$ — 1 mark $\therefore g = \frac{6.67 \times 10^{-11} \times 1.9 \times 10^{27}}{7.1 \times 10^7 \times 7.1 \times 10^7}$ $\therefore g = 25 \text{ m/s}^2$ (approx.) — 1 mark	1 1				
10	) Gravitational force between two bodies, $F \propto m_1 m_2$ — (1) $F \propto \frac{1}{d^2}$ — (2) } — 1 mark from eq (1) & (2), we get, $F \propto \frac{m_1 m_2}{d^2}$ <u>$\therefore F = G m_1 m_2 / d^2$</u> — 1 mark	1 1				

11	$m_1 s_1 (t_1 - t) = m_2 s_2 (t - t_2)$ $0.8 \times s_1 (10 - 20) = 1.5 \times 4200 (20 - 15)$ $s_1 = 492.187 \text{ J/kg}^\circ\text{C}$		1
12	- Anomalous expansion of water: The volume of water decreases instead of increasing when it is heated from 0°C to 4°C. - Water pipes burst in very cold winter or the survival of aquatic organisms below the ice surface on ponds, lake, etc.		1
13	a. Decomposition Reaction 1. a reactant breaks down into two or more products. 2. It occurs in the presence of certain conditions such as heat, pressure, etc.	b. Single displacement reaction 1. an atom or radical of one reactant is displaced by an atom or radical of another reactant. 2. It occurs by simple mixing or contact.	1
14	- $v = Hd$ - This graph is related to the study of the expansion of the universe, known as Hubble's Law.		1
15	A. Myopia 1. far objects cannot be seen in this defect. 2. It occurs due to decrease in focal length of eyelens.	B. Hypermetropia 1. near objects cannot be seen in this defect. 2. It occurs due to increase in focal length of eyelens.	1
16	- Y = stable and less reactive - X = double/triple covalent bond		1
खण्ड "घ" Section "D"			
17	(a) phishing, online financial fraud, etc. (b) – (i) Regularly follow and check official notices from the Cyber Bureau or police (ii) Never share private photos, videos, or sensitive personal information with strangers or unknown accounts online (1 mark for each correct answer) (iii) Mental and emotional distress, loss of privacy and security, social and reputational damage. (1 mark for each correct answer)		1
18	(a) F_2 (b) RR or rr (c) 1: 2: 1 (d) Law of Segregation states that during the formation of gametes, the two alleles for a specific trait separate (segregate) from each other so that each gamete carries only one allele for each gene.		1

19	(a) C (b) B (c) D (d) A	1 1 1 1				
20	(a)  (b) Diamond Sparkle: Diamonds have a very high refractive index and a very small critical angle (about 24.4°). This causes light entering the diamond to undergo multiple total internal reflections before escaping, creating a brilliant sparkle. (c) Two Applications: Optical fiber cables (used in high-speed internet) and endoscopes (used in medical surgery) (1 mark for each correct answer)	1 2				
21	(a) Here $V_p = 750V$ $V_s = \frac{750}{10} V = 75V$ $N_p = 400 \text{ turns}$ $N_s = ?$ Using formula, $\frac{V_s}{V_p} = \frac{N_s}{N_p} \quad (1 \text{ mark})$ $\text{or, } \frac{75}{750} = \frac{N_s}{400}$ $\text{or, } N_s = 40 \text{ turns.} \quad (1 \text{ mark})$ (b) The output voltage will be equal to the input voltage (750 V). (1 mark) (c) The output voltage will be 0 V. Direct current (D.C.) provides a constant magnetic field and cannot induce a voltage in the secondary coil. (1 mark)	1 2				
22	<table> <tr> <th>s-block elements</th> <th>p-block elements</th> </tr> <tr> <td> 1. In s-block elements, the last electron enters the s-subshell. 2. s-block elements are all metals (except Hydrogen). </td> <td> 1. In p-block elements, the last electron enters the p-subshell. 2. p-block contains a mix of metals, metalloids, and non-metals. </td> </tr> </table>	s-block elements	p-block elements	1. In s-block elements, the last electron enters the s-subshell. 2. s-block elements are all metals (except Hydrogen).	1. In p-block elements, the last electron enters the p-subshell. 2. p-block contains a mix of metals, metalloids, and non-metals.	1 1
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	(b) $2\text{Na} + \text{F}_2 \longrightarrow 2\text{NaF}$	2
23	a) $\text{CaCO}_3 + 2\text{HCl} \longrightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$ (b) CO_2 is heavier (denser) than air. (c) CO_2 is used for extinguishing fire. (d) Its overproduction increases the greenhouse effect, leading to global warming and climate change.	1 1 1 1